

# THE CHAIN RULE

**Note:** The composite function  $(f \circ g)$  is defined by  $(f \circ g)(x) = f(g(x))$

**Example 1:** Suppose  $f(x) = \sqrt{x}$  and  $g(x) = x - 2$

a) Express  $(f \circ g)(x)$

c) Express  $(g \circ f)(x)$

b) Evaluate  $f(g(4))$

d) Evaluate  $g(f(4))$

**Example 2:** Differentiate the function  $h(x) = (2x^3 + 5x)^{35}$ .

**Note:**

If we let  $f(u) = u^{35}$  and we let  $u = g(x) = 2x^3 + 5x$  then  $h(x) = f(g(x))$ .

Since we know how to differentiate  $f(u)$  and  $g(x)$  individually, the **chain rule** allows us to differentiate  $y = f(g(x))$ .

It turns out that the derivative of the composite function,  $h(x) = f(g(x))$  is the product of the derivatives of  $f(u)$  and  $g(x)$ .

That is,  $h'(x) = f'(u)g'(x)$ .

**Recall:**

$\frac{dy}{du}$  is the rate of change of  $y$  with respect to  $u$ .

$\frac{du}{dx}$  is the rate of change of  $u$  with respect to  $x$ .

$\frac{dy}{dx}$  is the rate of change of  $y$  with respect to  $x$ .

If  $g(x)$  is differentiable at  $x$  and  $f(g(x))$  is differentiable at  $g(x)$ , then the composite function,  $h(x) = f(g(x))$  or  $h(x) = (f \circ g)(x)$  is differentiable at  $x$  and  $h'(x)$  is given by the product  $h'(x) = f'(g(x))g'(x)$ .

In Leibniz notation, If  $y=f(u)$  and  $u=g(x)$ , are both differentiable functions, then

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}.$$

### Examples:

1. If  $y = 2u^2 + 6u$ ,  $u = 2 + v^3$ , and  $v = \sqrt{x}$  then evaluate  $\frac{dy}{dx}$  at  $x = 4$ .

2. Determine the derivative of  $f(x) = \sqrt{\frac{x-3}{3-5x}}$

3. Differentiate:

a)  $f(x) = (4x^2 + 3x)^5$

b)  $f(x) = (5x^2 - x)^x$

c)  $y = \frac{x}{1 - 4x^2}$

d)  $f(t) = (2t^3 + 6t - 5)^5$

e)  $y = (3x^2 - 5)\sqrt{2x^2 + 9x}$

f)  $m(t) = \sqrt[3]{1 + t + t^2}$

4. a) If  $h(x) = \frac{(f(x))^2}{g(x)}$ , determine  $h'(x)$ .

b) Given  $f(1) = 2$ ,  $f'(1) = -3$ ,  $g(1) = 1$  and  $g'(1) = 4$ , determine  $h'(1)$ .